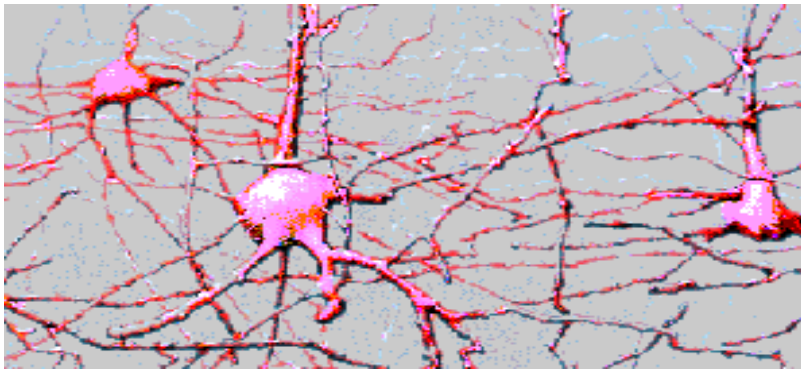
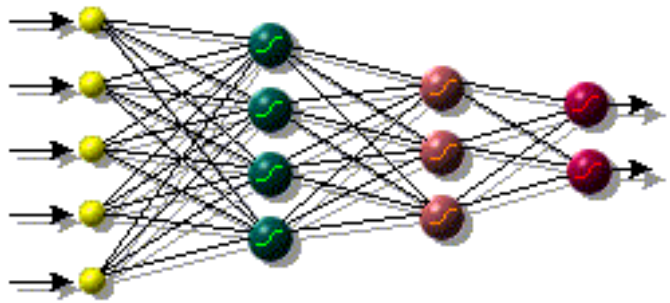
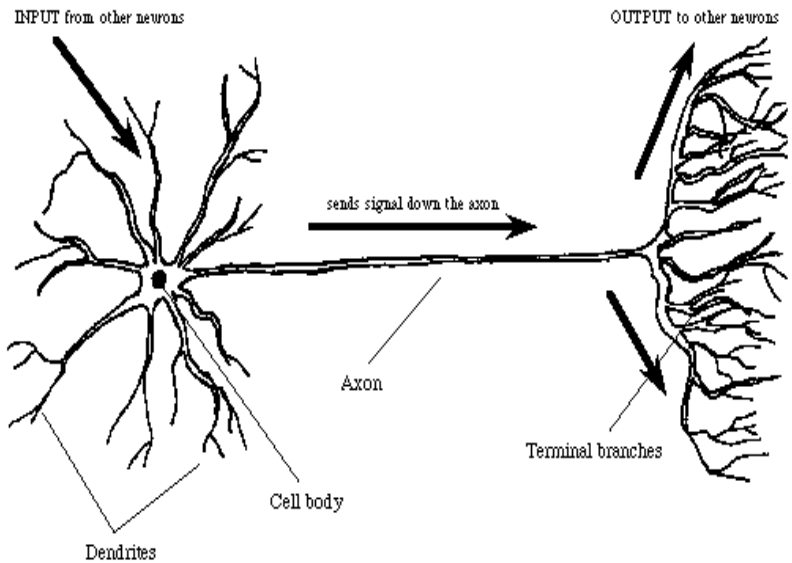
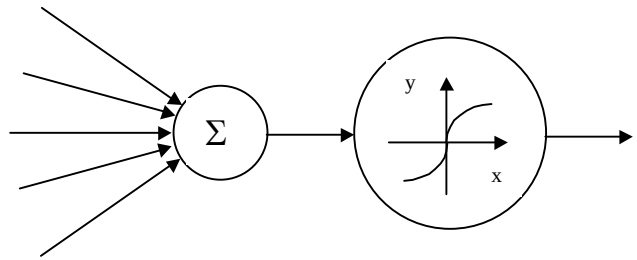




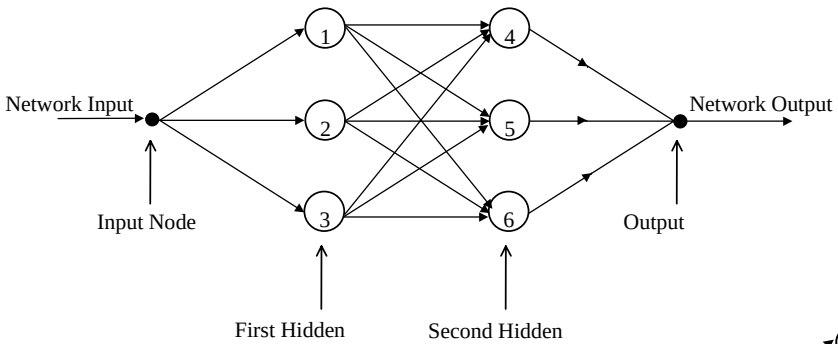
Artificial Neural Networks



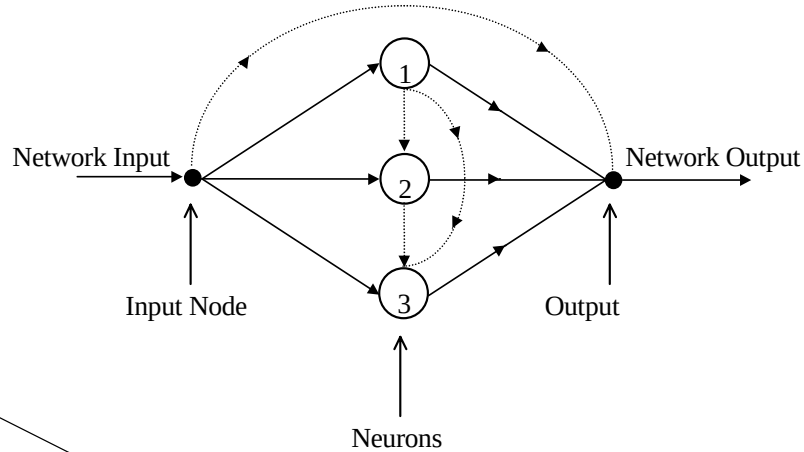


NN Architectures

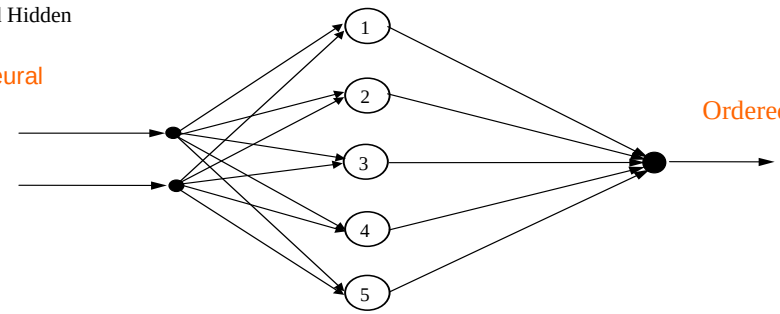
- *Artificial Neural Networks* are multi-input-multi-output systems composed of many inter-connected nonlinear processing elements (neurons) operating in parallel



Two-Hidden-Layers Neural Network



Ordered Neural Network



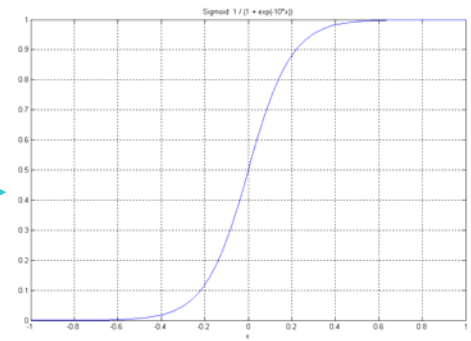
Single-Hidden-Layer Neural Network

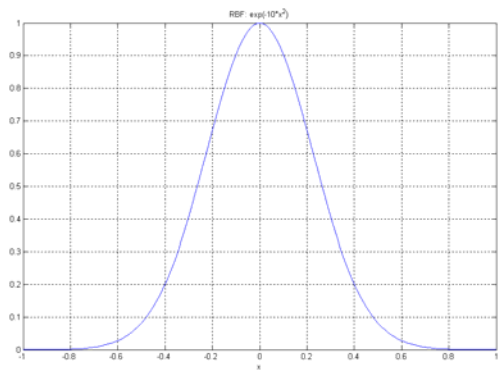
Single Hidden Layer (SHL) Feedforward Neural Networks (FNN)

- Three distinct characteristics
  - model of each neuron includes a ***nonlinear*** activation function
    - sigmoid
    - radial basis function
  - a single layer of N ***hidden*** neurons
  - ***feedforward*** connectivity

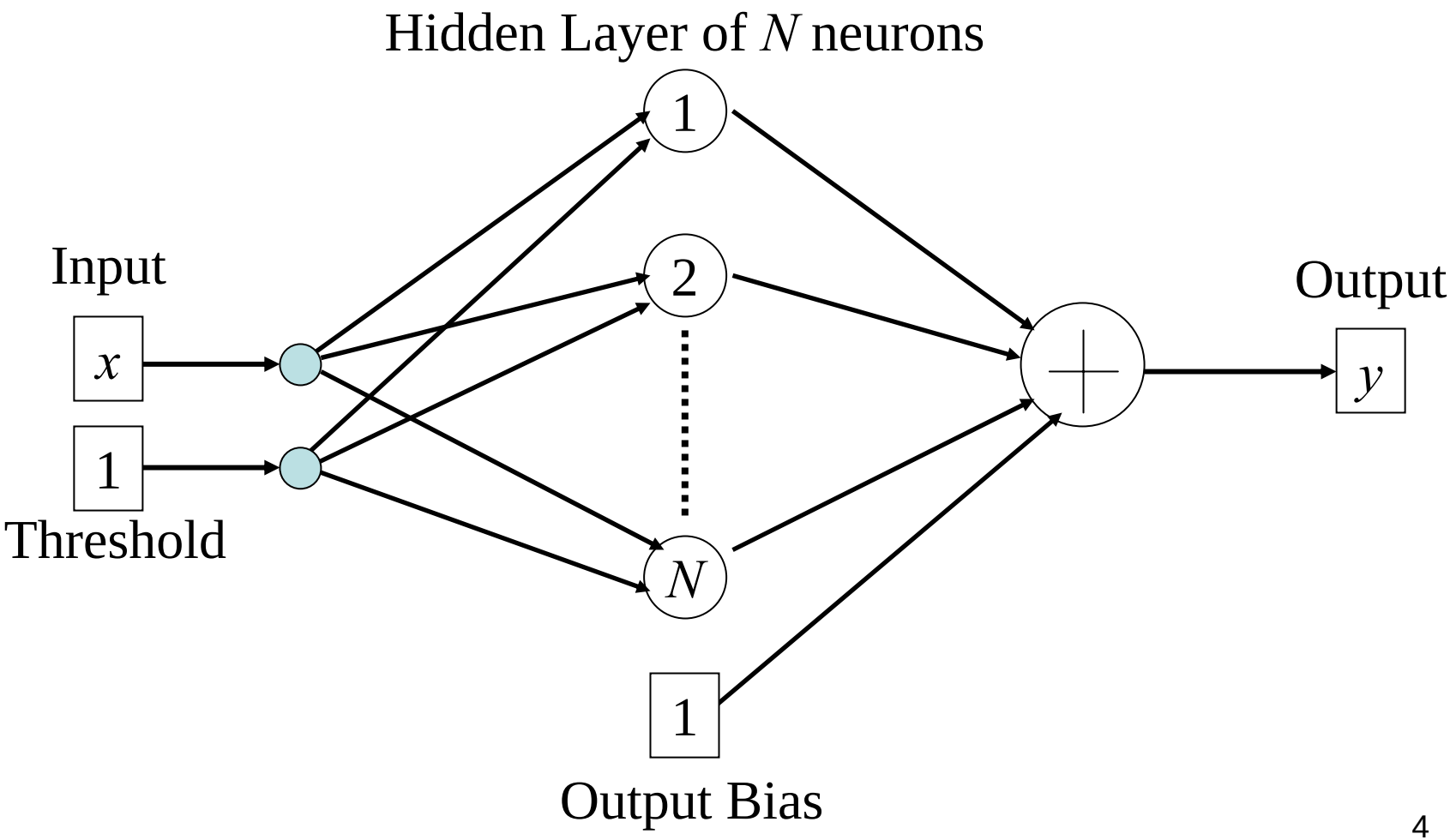
$$\sigma(s) = \frac{1}{1+e^{-s}}$$

$$\varphi(x) = e^{-\frac{\|x-r\|^2}{2\sigma^2}}$$





SHL FNN Architecture



SHL FNN Function

- Maps n - dimensional input into m - dimensional output: $x \to NN(x), \quad x \in R^n, \quad NN(x) \in R^m$
- Functional Dependence
  - sigmoidal: $NN(x) = W^T \vec{\sigma}(V^T x + \theta) + b$
  - RBF:

$$NN(x) = W^T \underbrace{\begin{pmatrix} \varphi(\|x - C_1\|) \\ \vdots \\ \varphi(\|x - C_N\|) \end{pmatrix}}_{\Phi(x)} + b = W^T \Phi(x) + b$$

Sigmoidal NN

- Matrix form: $$NN(x) = W^T \vec{\sigma}\left(V^T \begin{pmatrix} x \\ 1 \end{pmatrix} \right) + c$$
- Vector of hidden layer sigmoids: $$\vec{\sigma}\left(V^T x + \theta\right) = \left(\sigma\left(v_1^T x + \theta_1\right) \quad \ldots \quad \sigma\left(\vec{v}_N^T x + \theta_N\right)\right)^T$$
- Matrix of inner-layer weights: $$V = \left(\vec{v}_1 \quad \ldots \quad \vec{v}_N\right) \in R^{n \times N}$$
- Matrix of output-layer weights: $$W = \left(\vec{w}_1 \quad \ldots \quad \vec{w}_m\right) \in R^{N \times m}$$
- Vector of output biases $c \in R^m$ and thresholds $\theta \in R^N$
- k^{th} output: $$NN_k(x) = \vec{w}_k^T \sigma\left(\vec{v}_k^T x + \theta_k\right) + c_k = \sum_{j=1}^{N} w_{jk} \sigma\left(\sum_{i=1}^{n} v_{ik} x_i + \theta_k\right) + c_k$$

Sigmoidal NN, (continued)

- Universal Approximation Property
  - large class of functions can be approximated by sigmoidal SHL NN-s within any given tolerance, on compacted domains

$$\forall f(x): R^n \to R^m \quad \forall \varepsilon > 0 \quad \exists N, W, b, V, \theta \quad \forall x \in X \subset R^n$$

$$\left\| f(x) - W^T \vec{\sigma}\left(V^T \begin{pmatrix} x \\ 1 \end{pmatrix} \right) - b \right\| \leq \varepsilon = \mathrm{O}\left(\frac{1}{\sqrt{N}} \right)$$

- Introduce: $W \triangleq \begin{bmatrix} W^T & b \end{bmatrix}^T, \quad V \triangleq \begin{bmatrix} V^T & \theta \end{bmatrix}^T, \quad \vec{\sigma} \triangleq \begin{bmatrix} \vec{\sigma} \\ 1 \end{bmatrix}, \quad \mu \triangleq \begin{bmatrix} x \\ 1 \end{bmatrix}$
- Then: ⟹ $NN(x) = W^T \vec{\sigma}\left(V^T \mu \right)$

Sigmoidal SHL NN: Summary

- A very large class of functions can be approximated using ***linear combinations of shifted and scaled sigmoids***
- NN approximation error decreases as the number N of hidden-layer neurons increases:

$$\left\| f(x) - NN(x) \right\| = \mathrm{O}\left(N^{-\frac{1}{2}} \right)$$

- Inclusion of biases and thresholds into NN weight matrices simplifies bookkeeping

$$NN(x) = W^T \, \vec{\sigma}\left(V^T \, \mu \right)$$

- Function approximation using sigmoidal NN means finding connection weights W and V

RBF NN

- Matrix form: $NN(x) = W^T \Phi(x) + b$
- Vector of RBF-s:  $\Phi(x) = \begin{pmatrix} e^{\frac{-\|x-C_1\|^2}{2\sigma_1^2}} & \dots & e^{\frac{-\|x-C_N\|^2}{2\sigma_N^2}} \end{pmatrix}^T$
- Matrix of RBF centers: $C \triangleq \begin{bmatrix} \vec{C}_1 & \dots & \vec{C}_N \end{bmatrix} \in R^{n \times N}$
- Vector of RBF widths: $\vec{\sigma} \triangleq \begin{pmatrix} \sigma_1 & \dots & \sigma_N \end{pmatrix}^T \in R^N$
- Matrix of output weights: $W = \begin{pmatrix} \vec{w}_1 & \dots & \vec{w}_m \end{pmatrix} \in R^{N \times m}$
- Vector of output biases: $b \in R^m$
- k^{th} output: $NN_k(x) = \vec{w}_k^T \Phi(x) + b_k = \sum_{j=1}^{N} w_{jk} e^{\frac{-\|x-C_j\|^2}{2\sigma_j^2}} + b_k$

RBF NN, (continued)

- Universal Approximation Property
  - large class of functions can be approximated by RBF NN-s within any given tolerance, on compacted domains

$$\forall f(x): R^n \to R^m \quad \forall \varepsilon > 0 \quad \exists N, W, \vec{C}, \vec{\sigma} \quad \forall x \in X \subset R^n$$

$$\left\| f(x) - W^T \Phi(x) - b \right\| \leq \varepsilon = \mathrm{O}\left(N^{-\frac{1}{n}} \right)$$

- Introduce: $W \triangleq [W \quad b], \quad \Phi(x) \triangleq \begin{bmatrix} \Phi(x) \\ 1 \end{bmatrix}$
- Then: $NN(x) = W^T \Phi(x)$

RBF NN: Summary

- A very large class of functions can be approximated using *linear combinations of shifted and scaled gaussians*
- NN approximation error decreases as the number N of hidden-layer neurons increases:

$$\left\| f(x) - NN(x) \right\| = \mathrm{O}\left(N^{-\frac{1}{n}} \right)$$

- Inclusion of biases into NN output weight matrix simplifies bookkeeping

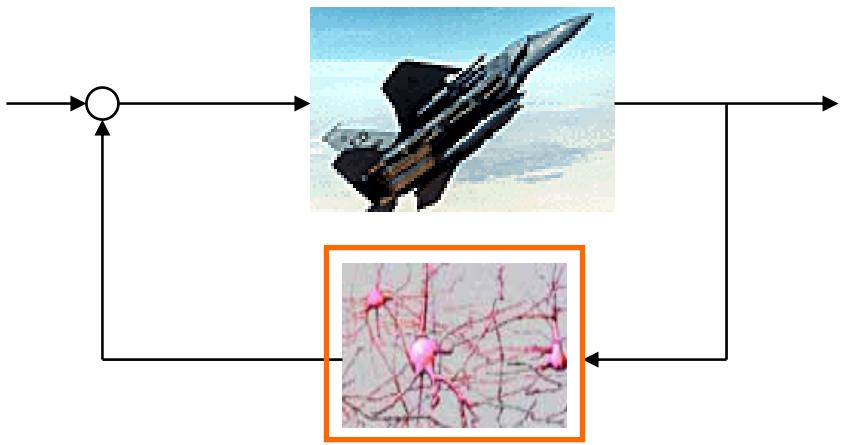
$$NN(x) = W^T \Phi(x)$$

- Function approximation using RBF NN means finding output weights W, centers C, and widths $\vec{\sigma}$

What is Next?

- Use SHL FNN-s in the context of MRAC systems
  - off-line / on-line approximation of uncertain nonlinearities in system dynamics
    - modeling errors, (aerodynamics)
    - damage
    - control failures
- Start with fixed widths RBF NN architectures, (linear in unknown parameters)
- Generalize to using sigmoidal NN-s

Adaptive NeuroControl



n^{th} Order Systems with Matched Uncertainties

- System Dynamics: $\dot{x} = A\,x + B\,\Lambda\left(u - f\left(x\right)\right), \quad x \in R^n, \quad u \in R^m$
  - $A \in R^{n\times n}, \Lambda = \text{diag}\left(\lambda_1 \quad \ldots \quad \lambda_m\right) \in R^{m\times m}$ are constant ***unknown*** matrices
  - $B \in R^{n\times m}$ is ***known*** constant matrix
  - $\forall i = 1,\ldots,m \quad \text{sgn}\left(\lambda_i\right)$ is ***known***
- ***Approximation of uncertainty***: $f\left(x\right) = \Theta^T\,\Phi\left(x\right) + \varepsilon_f\left(x\right)$
  - ***matrix*** of constant ***unknown*** parameters: $\Theta \in R^{m\times N}$
  - vector of N ***fixed*** RBF-s: $\Phi\left(x\right) = \left(\varphi_1\left(x\right) \quad \ldots \quad \varphi_N\left(x\right)\right)^T$
  - function approximation tolerance: $\varepsilon_f\left(x\right) \in R^m$

n^{th} Order Systems with Matched Uncertainties, (continued)

- *Assumption*: Number of RBF-s, true (unknown) output weights Θ and widths/ centers are such that RBF NN approximates the nonlinearity within given tolerance:

$$\left\| \varepsilon_f(x) \right\| = \left\| f(x) - \Theta^T \Phi(x) \right\| \le \varepsilon, \quad \forall x \in X \subset R^n$$

- RBF NN estimator: $\hat{f}(x) = \hat{\Theta}^T \Phi(x)$
- Estimation error:

$$NN(x) - f(x) = \underbrace{\left(\hat{\Theta} - \Theta\right)}_{\Delta\Theta}^T \Phi(x) - \varepsilon_f(x) = \Delta\Theta^T \Phi(x) - \varepsilon_f(x)$$

n^{th} Order Systems with Matched Uncertainties, (continued)

- Stable Reference Model: $\dot{x}_m = A_m x_m + B_m r, \quad (A_m \text{ is Hurwitz})$

$$r \in R^m, \quad A_m \in R^{n \times n}, \quad B_m \in R^{m \times m}$$

- **<u>Control Goal</u>**
  - bounded tracking: $\lim_{t \to \infty} \left\| x(t) - x_m(t) \right\| \leq \varepsilon_x$
- MRAC Design Process
  - choose N and vector of widths $\vec{\sigma}$
    - can be performed off-line in order to incorporate any prior knowledge about the uncertainty
  - design MRAC and evaluate closed-loop system performance
  - repeat previous two steps, if required

n^{th} Order Systems with Matched Uncertainties, (continued)

- Control Feedback: $u = \hat{K}_x^T\, x + \hat{K}_r^T\, r + \hat{\Theta}^T\, \Phi(x)$
  - $(m\, n + m^2 + m\, N)$ - parameters to estimate: $\hat{K}_x, \quad \hat{K}_r, \quad \hat{\Theta}$
- Closed-Loop: $\dot{x} = \left(A + B\,\Lambda\,\hat{K}_x^T\right) x + B\Lambda\left(\hat{K}_r^T\, r + \Delta\Theta^T\, \Phi(x) - \varepsilon_f(x)\right)$
- Desired Dynamics: $\dot{x}_m = A_m\, x_m + B_m\, r$
- Model Matching Conditions
  - there exist *ideal* gains (K_x, K_r) such that: $\Rightarrow$ $A + B\,\Lambda\,K_x^T = A_m$, $B\,\Lambda\,K_r^T = B_m$
  - *Note*: knowledge of the ideal gains is not required

$$A + B\,\Lambda\,\hat{K}_x^T - A_m = A + B\,\Lambda\,\hat{K}_x^T - A - B\,\Lambda\,K_x^T = B\,\Lambda\left(\hat{K}_x - K_x\right)^T = B\,\Lambda\,\Delta K_x^T$$

$$B\,\Lambda\,\hat{K}_r^T - B_m = B\,\Lambda\,\hat{K}_r^T - B\,\Lambda\,K_r^T = B\,\Lambda\left(\hat{K}_r - K_r\right)^T = B\,\Lambda\,\Delta\hat{K}_r^T$$

n^{th} Order Systems with Matched Uncertainties, (continued)

- Tracking Error: $\boxed{e(t) = x(t) - x_m(t)}$
- Error Dynamics:

$$\dot{e}(t) = \dot{x}(t) - \dot{x}_m(t) =$$

$$\left(A + B\Lambda \hat{K}_x^T\right)x + B\Lambda\left(\hat{K}_r^T r + \Delta\Theta^T \Phi(x) - \varepsilon_f(x)\right) - A_m x_m - B_m r \pm A_m x$$

$$= A_m\left(x - x_m\right) + \left(A + B\Lambda \hat{K}_x^T - A_m\right)x + B\Lambda\left(\hat{K}_r - K_r\right)^T r + B\Lambda\left(\Delta\Theta^T \Phi(x) - \varepsilon_f(x)\right)$$

$$= A_m e + B\Lambda\left(\Delta K_x^T x + \Delta K_r^T r + \Delta\Theta^T \Phi(x) - \varepsilon_f(x)\right)$$

- Remarks
  - estimation error $\varepsilon_f(x)$ is bounded, as long as $x \in X$
  - need to keep x within X

n^{th} Order Systems with Matched Uncertainties, (continued)

- Lyapunov Function Candidate

$$V\left(e,\Delta K_x,\Delta K_r,\Delta\Theta\right)=e^T P e$$
$$+\text{trace}\left(\Delta K_x^T\,\Gamma_x^{-1}\,\Delta K_x\left|\Lambda\right|\right)+\text{trace}\left(\Delta K_r^T\,\Gamma_r^{-1}\,\Delta K_r\left|\Lambda\right|\right)+\text{trace}\left(\Delta\Theta^T\,\Gamma_\Theta^{-1}\,\Delta\Theta\left|\Lambda\right|\right)$$

  - where: $\text{trace}\left(S\right)\triangleq\sum s_{ii}$
  - $\left|\Lambda\right|\triangleq\text{diag}\left(\left|\lambda_1\right|\quad\ldots\quad\left|\lambda_m^i\right|\right)$ is diagonal matrix with positive elements
  - $\Gamma_x=\Gamma_x^T>0,\quad\Gamma_r=\Gamma_r^T>0,\quad\Gamma_\Theta=\Gamma_\Theta^T>0$ are symmetric positive definite matrices
  - $P=P^T>0$ is unique symmetric positive definite solution of the algebraic Lyapunov equation $P\,A_m+A_m^T\,P=-Q$
    - $Q=Q^T>0$ is any symmetric positive definite matrix

n^{th} Order Systems with Matched Uncertainties, (continued)

- Time-derivative of the Lyapunov function

$$\dot{V} = \dot{e}^T P e + e^T P \dot{e}$$

$$+2\,\mathrm{trace}\left(\Delta K_x^T \Gamma_x^{-1} \dot{\hat{K}}_x |\Lambda|\right) + 2\,\mathrm{trace}\left(\Delta K_r^T \Gamma_r^{-1} \dot{\hat{K}}_r |\Lambda|\right) + 2\,\mathrm{trace}\left(\Delta\Theta^T \Gamma_\Theta^{-1} \dot{\hat{\Theta}} |\Lambda|\right)$$

$$= \left(A_m e + B\Lambda\left(\Delta K_x^T x + \Delta K_r^T r + \Delta\Theta^T \Phi(x) - \varepsilon_f(x)\right)\right)^T P e$$

$$+ e^T P \left(A_m e + B\Lambda\left(\Delta K_x^T x + \Delta K_r^T r + \Delta\Theta^T \Phi(x) - \varepsilon_f(x)\right)\right)$$

$$+2\,\mathrm{trace}\left(\Delta K_x^T \Gamma_x^{-1} \dot{\hat{K}}_x |\Lambda|\right) + 2\,\mathrm{trace}\left(\Delta K_r^T \Gamma_r^{-1} \dot{\hat{K}}_r |\Lambda|\right) + 2\,\mathrm{trace}\left(\Delta\Theta^T \Gamma_\Theta^{-1} \dot{\hat{\Theta}} |\Lambda|\right)$$

$$= e^T \left(A_m P + P A_m\right) e$$

$$+2 e^T P B\Lambda\left(\Delta K_x^T x + \Delta K_r^T r + \Delta\Theta^T \Phi(x) - \varepsilon_f(x)\right)$$

$$+2\,\mathrm{trace}\left(\Delta K_x^T \Gamma_x^{-1} \dot{\hat{K}}_x |\Lambda|\right) + 2\,\mathrm{trace}\left(\Delta K_r^T \Gamma_r^{-1} \dot{\hat{K}}_r |\Lambda|\right) + 2\,\mathrm{trace}\left(\Delta\Theta^T \Gamma_\Theta^{-1} \dot{\hat{\Theta}} |\Lambda|\right)$$

n^{th} Order Systems with Matched Uncertainties, (continued)

- Time-derivative of the Lyapunov function

$$\dot{V} = -e^T Q e - 2 e^T P B \Lambda \varepsilon_f(x)$$
$$+2 e^T P B \Lambda \Delta K_x^T x + 2\,\mathrm{trace}\left(\Delta K_x^T \Gamma_x^{-1} \dot{\hat{K}}_x |\Lambda|\right)$$
$$+2 e^T P B \Lambda \Delta K_r^T r + 2\,\mathrm{trace}\left(\Delta K_r^T \Gamma_r^{-1} \dot{\hat{K}}_r |\Lambda|\right)$$
$$+2 e^T P B \Lambda \Delta \Theta^T \Phi(x) + 2\,\mathrm{trace}\left(\Delta \Theta^T \Gamma_\Theta^{-1} \dot{\hat{\Theta}} |\Lambda|\right)$$

- Using trace identity: $a^T b = \mathrm{trace}\left(b a^T\right)$
- Example: $\underbrace{e^T P B \Lambda}_{a^T} \underbrace{\Delta K_x^T x}_{b} = \mathrm{trace}\left(\underbrace{\Delta K_x^T x}_{b} \underbrace{e^T P B \Lambda}_{a^T}\right)$

n^{th} Order Systems with Matched Uncertainties, (continued)

- Time-derivative of the Lyapunov function

$$\dot{V} = -e^T Q e - 2 e^T P B \Lambda \varepsilon_f(x)$$
$$+2\,\text{trace}\left(\Delta K_x^T \left\{ \Gamma_x^{-1} \dot{\hat{K}}_x + x e^T P B \,\text{sgn}(\Lambda) \right\} |\Lambda| \right)$$
$$+2\,\text{trace}\left(\Delta K_r^T \left\{ \Gamma_r^{-1} \dot{\hat{K}}_r + r e^T P B \,\text{sgn}(\Lambda) \right\} |\Lambda| \right)$$
$$+2\,\text{trace}\left(\Delta \Theta^T \left\{ \Gamma_\Theta^{-1} \dot{\hat{\Theta}} + \Phi(x) e^T P B \,\text{sgn}(\Lambda) \right\} |\Lambda| \right)$$

- Problem
  - choose adaptive parameters $\hat{K}_x, \hat{K}_r, \hat{\Theta}$ such that time-derivative $\dot{V}$ becomes negative definite outside of a compact set in the *error* state space, and all parameters remain bounded for all future times

n^{th} Order Systems with Matched Uncertainties, (continued)

- Suppose that we choose adaptive laws:

$$\dot{\hat{K}}_x = -\Gamma_x\, x\, e^T P B \operatorname{sgn}(\Lambda)$$

$$\dot{\hat{K}}_r = -\Gamma_r\, r\, e^T P B \operatorname{sgn}(\Lambda)$$

$$\dot{\hat{\Theta}} = -\Gamma_\Theta\, \Phi(x)\, e^T P B \operatorname{sgn}(\Lambda)$$

- Then we get:

$$\dot{V} = -e^T Q e - 2 e^T P B \Lambda \varepsilon_f(x) \le -\lambda_{\min}(Q)\|e\|^2 + 2\|e\|\|PB\|\lambda_{\max}(\Lambda)\varepsilon$$

- Consequently, $\dot{V} < 0$ outside of the set

$$E \triangleq \left\{ e : \|e\| \le \frac{2\|PB\|\lambda_{\max}(\Lambda)\varepsilon}{\lambda_{\min}(Q)} \right\}$$

- *Unfortunately*, inside E parameter errors may grow out of bounds, (for $e \in E$, $\dot{V}$ IS NOT necessarily negative!)

How to Keep Adaptive Parameters Bounded?

- σ - modification:

$$\dot{\hat{K}}_x = -\Gamma_x \left(x e^T P B + \sigma_x \hat{K}_x \right) \mathrm{sgn}(\Lambda)$$

$$\dot{\hat{K}}_r = -\Gamma_r \left(r e^T P B + \sigma_r \hat{K}_r \right) \mathrm{sgn}(\Lambda)$$

$$\dot{\hat{\Theta}} = -\Gamma_\Theta \left(\Phi(x) e^T P B + \sigma_\Theta \hat{\Theta} \right) \mathrm{sgn}(\Lambda)$$

- e - modification:

$$\dot{\hat{K}}_x = -\Gamma_x \left(x e^T P B + \sigma_x \left\| e^T P B \right\| \hat{K}_x \right) \mathrm{sgn}(\Lambda)$$

$$\dot{\hat{K}}_r = -\Gamma_r \left(r e^T P B + \sigma_r \left\| e^T P B \right\| \hat{K}_r \right) \mathrm{sgn}(\Lambda)$$

$$\dot{\hat{\Theta}} = -\Gamma_\Theta \left(\Phi(x) e^T P B + \sigma_\Theta \left\| e^T P B \right\| \hat{\Theta} \right) \mathrm{sgn}(\Lambda)$$

- Modifications add *damping* to adaptive laws
  - damping controlled by choosing $\sigma_x, \sigma_r, \sigma_\Theta > 0$
  - there is a *trade off* between adaptation rate and damping

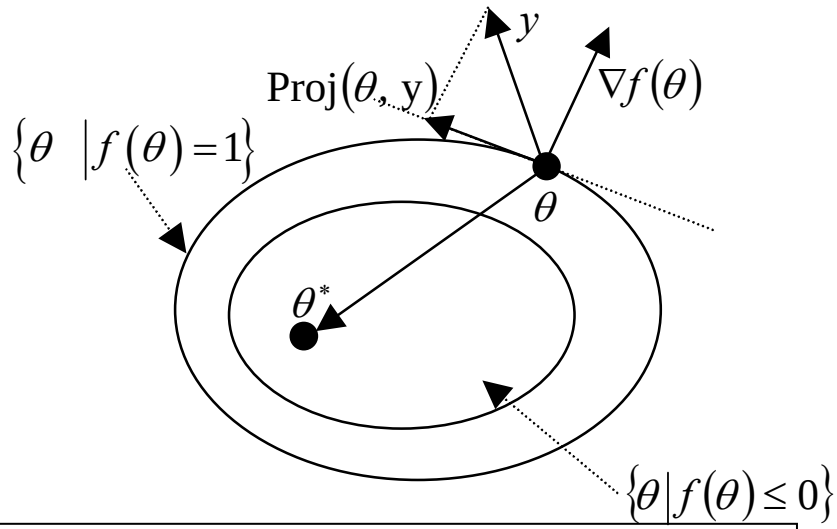
Introducing Projection Operator

- Requires no damping terms
- Designed to keep NN weights within *pre-specified* bounds
- Maintains negative values of the Lyapunov function time-derivative outside of subset: $$E \triangleq \left\{ e : \|e\| \leq \frac{2\|PB\|\lambda_{\max}(\Lambda)\varepsilon}{\lambda_{\min}(Q)} \right\}$$
  - the size of E defines tracking tolerance
  - the size of E can be controlled!

Projection Operator

- Function $f(\theta)$ defines pre-specified parameter domain boundary
- Example:

$$f(\theta) = \frac{\|\theta\|^2 - \theta_{\max}^2}{\varepsilon_\theta \, \theta_{\max}^2}$$



$$\{f(\theta) \le 0\} \Rightarrow \{\|\theta\| \le \theta_{\max}\} \Rightarrow \theta \text{ is within bounds}$$

$$\{0 < f(\theta) \le 1\} \Rightarrow \{\|\theta\| \le \sqrt{1+\varepsilon_\theta}\,\theta_{\max}\} \Rightarrow \theta \text{ is within } \left(\sqrt{1+\varepsilon_\theta}\right)\% \text{ of bounds}$$

$$\{f(\theta) > 1\} \Rightarrow \{\|\theta\| > \sqrt{1+\varepsilon_\theta}\,\theta_{\max}\} \Rightarrow \theta \text{ is outside of bounds}$$

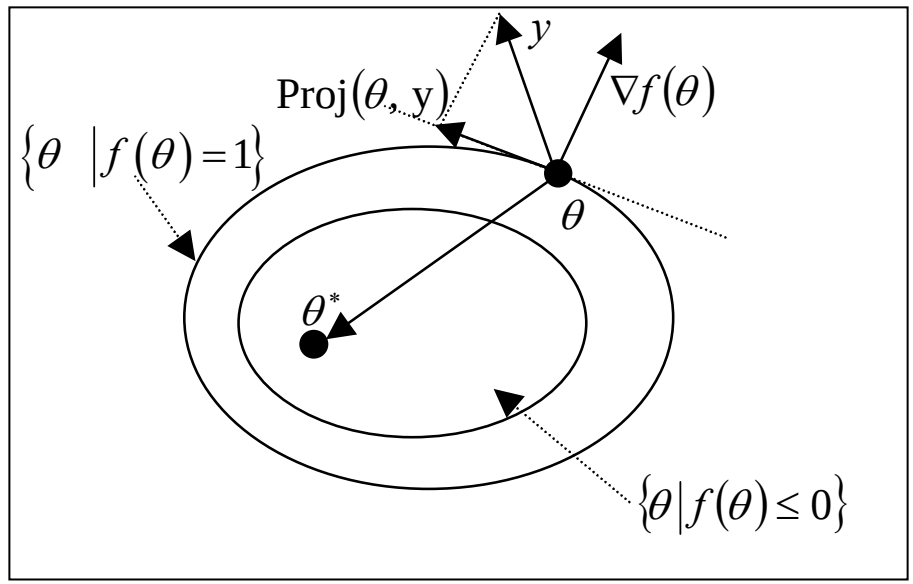
- $\theta_{\max}$ specifies boundary
- ε_θ specifies boundary tolerance

Projection Operator, (continued)

- ***Definition***:

$$\text{Proj}(\theta, y) = \begin{cases} y - \dfrac{\nabla f(\theta)\left(\nabla f(\theta)\right)^T}{\left\|\nabla f(\theta)\right\|^2} y\, f(\theta), & \text{if } f(\theta) > 0 \text{ and } y^T \nabla f(\theta) > 0 \\ y, & \text{if not} \end{cases}$$

- Depends on (θ, y)
- Does not alter y if θ is within the pre-specified bounds: $\|\theta\| \le \theta_{\max}$
- Gradient: $\nabla f(\theta) = \dfrac{2\theta}{\varepsilon_\theta \theta_{\max}^2}$
- In $\{0 \le f(\theta) \le 1\}$ the operator subtracts gradient vector $\nabla f(\theta)$ (normal to the boundary) from y
  - get a *smooth* transition from y for $\lambda = 0$ to a tangent vector field for $\lambda = 1$



- ***Important Property*** $\left(\theta - \theta^*\right)^T \left(\text{Proj}(\theta, y) - y\right) \le 0$

Lyapunov Function Time-Derivative with Projection Operator

- Make trace terms semi-negative *AND* keep parameters bounded:

$$\dot{V} = -e^T Q e - 2 e^T P B \Lambda \varepsilon_f(x)$$

$$+2\,\mathrm{trace}\left(\Delta K_x^T \left\{ \underbrace{\Gamma_x^{-1} \dot{\hat{K}}_x}_{\mathrm{Proj}\left(\hat{K}_x, y\right)} + \underbrace{x\, e^T P B \,\mathrm{sgn}(\Lambda)}_{-y} \right\} |\Lambda| \right)$$

$$+2\,\mathrm{trace}\left(\Delta K_r^T \left\{ \underbrace{\Gamma_r^{-1} \dot{\hat{K}}_r}_{\mathrm{Proj}\left(\hat{K}_r, y\right)} + \underbrace{r\, e^T P B \,\mathrm{sgn}(\Lambda)}_{-y} \right\} |\Lambda| \right)$$

$$+2\,\mathrm{trace}\left(\Delta \Theta^T \left\{ \underbrace{\Gamma_\Theta^{-1} \dot{\hat{\Theta}}}_{\mathrm{Proj}\left(\hat{\Theta}, y\right)} + \underbrace{\Phi(x) e^T P B \,\mathrm{sgn}(\Lambda)}_{-y} \right\} |\Lambda| \right)$$

Adaptation with Projection

- Modified adaptive laws:

$$\dot{\hat{K}}_x = \Gamma_x \,\mathrm{Proj}\left(\hat{K}_x, -x\, e^T P B \,\mathrm{sgn}\left(\Lambda\right)\right)$$

$$\dot{\hat{K}}_r = \Gamma_r \,\mathrm{Proj}\left(\hat{K}_r, -r\, e^T P B \,\mathrm{sgn}\left(\Lambda\right)\right)$$

$$\dot{\hat{\Theta}} = \Gamma_\Theta \,\mathrm{Proj}\left(\hat{\Theta}, -\Phi\left(x\right) e^T P B \,\mathrm{sgn}\left(\Lambda\right)\right)$$

- Projection Operator, its bounds and tolerances are defined *column-wise*
- Lyapunov function time-derivative:

$$\dot{V} \le -e^T Q e - 2 e^T P B \Lambda \varepsilon_f\left(x\right) \le -\lambda_{\min}\left(Q\right)\left\|e\right\|^2 + 2\left\|e\right\|\left\|P B\right\| \lambda_{\max}\left(\Lambda\right)\varepsilon$$

- Adaptive parameters stay within the pre-specified bounds, while $\dot{V} < 0$ outside of the set:

$$E \triangleq \left\{ e : \left\|e\right\| \le \frac{2\left\|P B\right\| \lambda_{\max}\left(\Lambda\right)\varepsilon}{\lambda_{\min}\left(Q\right)} \right\}$$

Example: Projection Operator, (scalar case)

- Scalar adaptive gain: $\dot{\hat{k}} = \gamma \, \mathrm{Proj}\left(\hat{k}, -x\,e\,\mathrm{sgn}(b)\right)$
- Pre-specified parameter domain boundary:
  - using function: $f\left(\hat{k}\right) = \dfrac{\hat{k}^2 - k_{\max}^2}{\varepsilon\, k_{\max}^2}$ $\Rightarrow$ $\nabla f\left(\hat{k}\right) = f'\left(\hat{k}\right) = \dfrac{2\hat{k}}{\varepsilon\, k_{\max}^2}$

$$\left\{ f\left(\hat{k}\right) \le 0 \right\} \Rightarrow \left\{ \left|\hat{k}\right| \le k_{\max} \right\} \Rightarrow \hat{k} \text{ is within bounds}$$

$$\left\{ 0 < f\left(\hat{k}\right) \le 1 \right\} \Rightarrow \left\{ \left|\hat{k}\right| \le \sqrt{1+\varepsilon}\, k_{\max} \right\} \Rightarrow \hat{k} \text{ is within } \left(\sqrt{1+\varepsilon}\right)\% \text{ of bounds}$$

$$\left\{ f\left(\hat{k}\right) > 1 \right\} \Rightarrow \left\{ \left|\hat{k}\right| > \sqrt{1+\varepsilon}\, k_{\max} \right\} \Rightarrow \hat{k} \text{ is outside of bounds}$$

  - Projection Operator:

$y = -x\,e\,\mathrm{sgn}(b)$ $\Rightarrow$

$$\mathrm{Proj}\left(\hat{k}, y\right) = \begin{cases} y\left(1 - f\left(\hat{k}\right)\right), & \text{if } f\left(\hat{k}\right) > 0 \text{ and } y\, f'\left(\hat{k}\right) > 0 \\ y, & \text{if not} \end{cases}$$

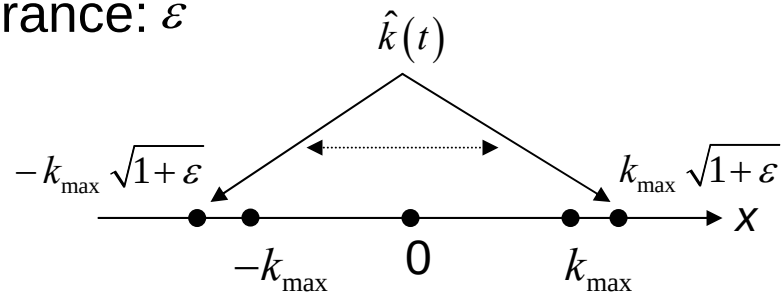
Example: Projection Operator, (scalar case) (continued)

- Adaptive Law, ($b > 0$):

$$\dot{\hat{k}} = \begin{cases} -x\,e\left(1 - f\left(\hat{k}\right)\right), & \text{if} \quad \left[f\left(\hat{k}\right)\right] > 0 \quad \text{and} \quad \left[x\,e\,f'\left(\hat{k}\right)\right] < 0 \\ -x\,e, & \text{if not} \end{cases}$$

$$\text{where:} \quad f\left(\hat{k}\right) = \frac{\hat{k}^2 - k_{\max}^2}{\varepsilon\, k_{\max}^2}$$

- Geometric Interpretation
  - adaptive parameter $\hat{k}(t)$ changes within the pre-specified interval
  - interval bound: $k_{\max}$
  - Bound tolerance: ε



Adaptive *Augmentation* Design

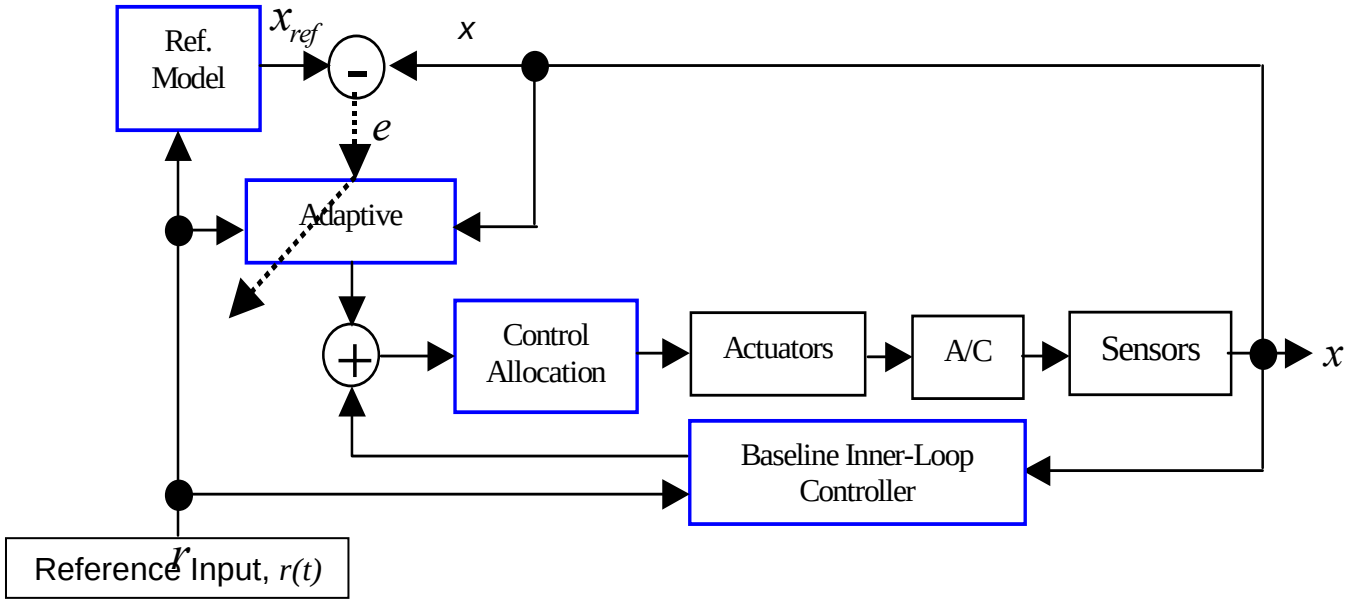
- Nominal Control: $u_{nom} = F_x^T\, x + F_r^T\, r$
- Adaptive Control: $u = \hat{K}_x^T\, x + \hat{K}_r^T\, r + \hat{\Theta}^T\, \Phi(x)$
- Augmentation:

$$
\begin{aligned}
u &= \hat{K}_x^T\, x + \hat{K}_r^T\, r + \hat{\Theta}^T\, \Phi(x) \pm u_{nom} \\
&= u_{nom} + \underbrace{\left(\hat{K}_x - F_x\right)^T}_{\hat{D}_x^T}\, x + \underbrace{\left(\hat{K}_r - F_r\right)^T}_{\hat{D}_r^T}\, r + \hat{\Theta}^T\, \Phi(x) \\
&= u_{nom} + \hat{D}_x^T\, x + \hat{D}_r^T\, x + \hat{\Theta}^T\, \Phi(x)
\end{aligned}
$$

- *Incremental* Adaptation:

$$
\begin{aligned}
\dot{\hat{D}}_x &= \Gamma_x\, \mathrm{Proj}\left(\hat{D}_x,\ -x\, e^T P B\, \mathrm{sgn}(\Lambda)\right), \quad \hat{D}_x = 0_{n\times m} \\
\dot{\hat{D}}_r &= \Gamma_r\, \mathrm{Proj}\left(\hat{D}_r,\ -r\, e^T P B\, \mathrm{sgn}(\Lambda)\right), \quad \hat{D}_r = 0_{m\times m} \\
\dot{\hat{\Theta}} &= \Gamma_\Theta\, \mathrm{Proj}\left(\hat{\Theta},\ -\Phi(x)\, e^T P B\, \mathrm{sgn}(\Lambda)\right), \quad \hat{\Theta} = 0_{N\times m}
\end{aligned}
$$

Adaptive *Augmentation* Block-Diagram



- Reference Model provides desired response
- Nominal Baseline Controller
- Adaptive Augmentation
  - *Dead-Zone* modification prevents adaptation from changing nominal closed-loop dynamics
  - *Projection Operator* bounds adaptation parameters / gains

Adaptive Control using *Sigmoidal* NN

- System Dynamics: $\dot{x} = A\,x + B\,\Lambda\left(u - f\left(x\right)\right), \quad x \in R^n, \quad u \in R^m$
  - $A \in R^{n\times n}, \Lambda = \mathrm{diag}\left(\lambda_1 \quad \dots \quad \lambda_m\right) \in R^{m\times m}$ are constant *unknown* matrices
  - $B \in R^{M\times m}$ is *known* constant matrix, and $M \geq m$
  - $\forall\, i = 1,\dots,m \quad \mathrm{sgn}\left(\lambda_i\right)$ is *known*
- ***Approximation of uncertainty*:**

$$f\left(x\right) = W^T\,\vec{\sigma}\left(V^T\,\mu\right) + \varepsilon_f\left(x\right), \quad \mu = \begin{pmatrix} x^T & 1 \end{pmatrix}^T, \quad \varepsilon_f\left(x\right) \in R^m$$

  - *matrix* of constant *unknown* *Inner-Layer* weights: $V = \begin{bmatrix} \vec{v}_1 & \dots & \vec{v}_N \\ \theta_1 & \dots & \theta_N \end{bmatrix} \in R^{(n+1)\times N}$
  - *matrix* of constant *unknown* *Outer-Layer* weights: $W = \begin{bmatrix} \vec{w}_1 & \dots & \vec{w}_m \\ c_1 & \dots & c_m \end{bmatrix} \in R^{(N+1)\times m}$
  - vector of N *sigmoids* and a unity:

$$\vec{\sigma}\left(V^T\mu\right) = \begin{pmatrix} \sigma\left(v_1^T\,x + \theta_1\right) & \dots & \sigma\left(\vec{v}_N^T\,x + \theta_N\right) & 1 \end{pmatrix}^T, \quad \text{where: } \sigma\left(s\right) = \frac{1}{1+e^{-s}}$$

Adaptive Control using *Sigmoidal* NN

- Control Feedback: $\boxed{u = \hat{K}_x^T\, x + \hat{K}_r^T\, r + \hat{W}^T\, \vec{\sigma}\left(\hat{V}^T\, \mu\right)}$
  - $(m\, n + m^2 + (n + 1)\, N + (N + 1)\, m)$ - parameters to estimate: $\hat{K}_x,\quad \hat{K}_r,\quad \hat{W},\quad \hat{V}$
- Adaptation with Projection, $(\Lambda > 0)$:

$$\begin{cases} \dot{\hat{K}}_x = \Gamma_x\, \mathrm{Proj}\left(\hat{K}_x, -x\, e^T\, P\, B\right) \\ \dot{\hat{K}}_u = \Gamma_u\, \mathrm{Proj}\left(\hat{K}_u, -r\, e^T\, P\, B\right) \\ \dot{\hat{W}} = \Gamma_W\, \mathrm{Proj}\left(\hat{W}, \left(\vec{\sigma}\left(\hat{V}^T\, \mu\right) - \vec{\sigma}'\left(\hat{V}^T \mu\right)\hat{V}^T\, \mu\right) e^T P\, B\right) \\ \dot{\hat{V}} = \Gamma_V\, \mathrm{Proj}\left(\hat{V},\, \mu\, e^T P\, B \hat{W}^T\, \vec{\sigma}'\left(\hat{V}^T \mu\right)\right) \end{cases}$$

- Provides ***bounded*** tracking